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The Cooperation of Humans and Killer Whales (*Orcinus orca*): The Application of a Simple Fuzzy Rule-Based Model to a Historical System

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Brief author biographies

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J. B. Owens is Professor of History (*emeritus*) and former director of the Geographically-Integrated History Laboratory at Idaho State University (USA). He is currently completing a book, *Precarious Existence and Trust in the Spanish Kingdom of Murcia, 1550-1580*, which narrates how politically important Spanish men and women shaped an enduring form of cooperation among themselves and with others in the face of *unpredicted and unpredictable* events. Owens has treated nonlinear dynamics and the political system in an earlier book, *"By My Absolute Royal Authority": Justice and the Castilian Commonwealth at the Beginning of the First Global Age* (2005), which argues that perceptions of royal judicial administration shaped the degree of collaboration with the Crown by the kingdom's politically important groups. Owens is also working on a book in which he explains the high levels of cooperation in smuggling networks during the period 1550-1570. Although goods were moved between Central Europe and the Americas, the book focuses primarily on the kingdoms of Castile and Valencia and the Duchy of Milan, which had the same ruler. This work has been supported by fellowships from the U.S. National Endowment for the Humanities (2004-2005) and the John Simon Guggenheim Memorial Foundation (2005-2006), as well as by several grants from Idaho State University.

Owensô interest in human cooperation and the use of geographic information systems (GIS) for historical research led him to create a multidisciplinary, multinational research project for the European Science Foundationôs EUROCORES (European Collaborative Research) Schemeôs program ñThe Evolution of Cooperation and Tradingò (TECT; 2007-2010). The title of his project was ñDynamic Complexity of Self-Organizing Cooperation-Based Commercial Networks in the First Global Ageò [acronym: DynCoopNet]. His participation was funded by the

U.S. National Science Foundation (NSF). Subsequently, Owens was the lead Principal Investigator (PI) for a collaborative research project entitled ñUnderstanding social networks within complex, nonlinear systems: geographically-integrated history and dynamics GISò [acronym: SOCNET], which is administered by the U.S. National Science Foundation (NSF) Office of Cyberinfrastructure (OCI). The award is part of NSFôs Cyber-Enabled Discovery and Innovation (CDI) program.

He has published two other books and numerous journal articles, book chapters, and reviews in the Americas, Asia, and Europe. In 2002, Owens was recognized as his universityôs ñDistinguished Researcherò for the centennial. He was named in 1998 as one of the first group of "Fellows of Early Modern Studies" by the Council of the Sixteenth Century Studies Conference "for significant contributions to the advancement of scholarship and learning." He has also been recognized for his promotion of civil and human rights.

Ferenc Szidarovszky is the vice president of NOAH, a consulting firm dealing mainly with water resources management. He is also a senior researcher with the Ridgetop Group Inc. working on theoretical issues concerning reliability, robustness, optimal maintenance and replacement policies and prognostics in complex systems. In addition to these industrial applications he also performs research in systems theory, especially in dynamic economic systems. He has earned two PhD degrees, one in Applied Mathematics and another in Economics in Hungary. The Hungarian Academy of Sciences awarded him two higher degrees, Candidate in Mathematics and Doctor of Engineering Sciences. After graduating in Hungary, he became a professor of the Eotvos University of Sciences, Budapest, and later an acting department head in the Budapest University of Horticulture and Food Industry. In 1987, he moved to the US, and from 1988, he was professor of the Systems and Industrial Engineering Department of the University of Arizona. From September 2011, he was a Senior Researcher in Reliasoft Corporation, and from July 2015, he is with Ridgetop Group. As a Hungarian professor, he was an investigator in the first US-Hungarian joint research project jointly supported by the NSF and Hungarian Academy of Sciences on Decision Models in Water Resources Management. He has performed research in very broad fields of applied mathematics, including numerical analysis, optimization, multiobjective programming, game theory, dynamic systems, applied probability and their applications in industrial processes, environment, water resources management, economics, and computer network security among others. He is the author of 20 books and over 350 articles in international journals, in addition to numerous conference presentations and short courses taught in many countries. He is also a professor in the Applied Mathematics Department of the University of Pecs, Hungary, where he teaches a one-week, intensive Game Theory Course every year.

Abstract

For about a century, until the late 1920s, killer whales (*Orcinus orca*) and shore-based human whalers, both Aboriginals and those of European descent, in southeastern Australia cooperated to hunt large baleen whales. As is often the case in historical research, the surviving sources provide us with only vague, imprecise, anecdotal, fragmented, and ambiguous accounts of this fascinating case of animal-human cooperation (or mutualism). In order to gain at least a partial understanding of this situation, as a pedagogical exercise for historians and historical social scientists, we explain the creation of an atemporal mathematical fuzzy rule-based model that helps us make sense of the information we have in order to enrich the historical narrative. We employ the strengths of fuzzy logic because information can be imprecise or vague and we can use variables that are subjective in nature (for example, *cooperation*). The rules on which the model is built are transparent and are thus readily understood and discussed among domain experts, thereby improving debates among historians. The modeling process remains open-ended so that in the future, we can include other models and/or variables/classes. Because fuzzy logic (fuzzy set theory) and mathematical modeling are well-established techniques for dealing with vague information with which historians are largely unfamiliar, we intend this article to serve a pedagogical purpose by teaching historians how to do this type of modeling work when their sources, particularly those that are qualitative, exhibit such vagueness.

Keywords

Cooperation; mutualism; fuzzy rule-based modeling; shore-based whaling; *Orcinus orca*; killer whale; Australian history; Twofold Bay (Eden, Australia)

For Lucien Duckstein (1932-2015): fuzzy logic expert (Bardossy and Duckstein 1995), Holocaust survivor (Kroh 1996), mentor and friend.

Historians offer ñNarrative Knowledgeò to provide a unique and valuable means for understanding some aspect of real-world processes. We are not the first to argue for the importance of model creation for the development of narratives (Cronin 1992; Panel 2011). Although we reject his radical postmodernist perspective, ecological theorist Timothy Allen and his collaborators have published several works that expose the importance of narrative for scientific understanding and the relationship of model creation to providing narrative knowledge (for example, Allen and Hoekstra 2015: chap. 8; Allen et al. 2005; Zellmer et al. 2006). In order to enhance the value of this type of abstraction in the articles and books of historians, modeling offers a process of thought and useful tools. Historians can employ an atemporal model productively to assess the meaning of fragments of information and establish their connections in order to develop a narrative that will provide an improved understanding of past actions, events, and processes. To further this end, we present an historical model application that is pedagogical in nature, in that it presents the methodology for constructing a simple fuzzy model for a vague but complicated social cooperative network along with example model-simulation results.

Our starting point for writing this article was a video, ñKillers in Edenò, which was shown as part of the series ñNatureò on PBS, the U.S. public broadcasting system (McKee, no date; Toft 2004). This segment presented the historical case of cooperation for at least a century among orcas (killer whales), Aboriginal whalers, and shore-based, bay whalers of European origin to hunt huge baleen whales (e.g. humpback whales, southern right whales) migrating past the coast of southeastern Australia. Bay whaling helped sustain European immigration to Australia in the nineteenth century, but by 1878, most of this type of whaling activity had disappeared from most places. In Twofold Bay, in New South Wales, cooperation between humans and orcas may have sustained bay whaling for a longer period. There were 27 small, open boats with Aboriginal and European crews engaged in this activity in Twofold Bay in the period 1845-1848. By the 1890s, only the Davidson family regularly continued to hunt in cooperation with orcas, and even this remnant of shore-based whaling disappeared in the 1920s. Seen from the larger perspective of colonial histories, this story of human and whale interaction expands our comprehension beyond accounts limited to human actors. For those engaged in ñcooperation studiesò or in trying to understand the history of whaling or of Australia, this case fascinates and demands explanation.[1]

In order to gain at least a partial understanding of this situation, we propose creating a model based on the fragmentary record that remains. This story forms a small part of an open, complex human system that was coupled with physical and biological systems. Our information about such large systems remains inherently *vague*. However, the computers that we increasingly use to manage information rely on numbers and demand precision (*crispness*). How can we move from our imprecise information to the necessary precision (for example, in order to enter

[1] On the significance of whaling for the movement and settlement of humans in the North Pacific, see Jones 2013a, 2013b, and 2014.

information from our sources into some sort of computer spreadsheet)? How can one squeeze into a single field or category the information learned from statements in our sources, such as ñAmy is prettyò or ñthe level of trust is very strongò? There are *linguistic rules* that we can devise to describe a historical situation and to promote discussion about its characteristics. These rules can be transformed to represent the inherent *fuzziness* of the interactions and their larger environmental context. For a given situation, we *combine the fuzzy rules proportionately* to the degree to which they apply. We then *defuzzify* to obtain a *crisp* answer. The fuzzy rules represent the dynamic behavior of interest. Each fuzzy rule typically consists of an ñif-thenò premise. The ñifò connects the variables or factors discovered in the historical sources or through analogy, which determine the response or output. The ñthenò is the output variable. By describing for historians the development of this type of model, we will provide them with an important way to improve their approximations of the information that can be extracted from the surviving, always incomplete historical record of open, human-centered systems. Historians will then be able to use with greater profit other valuable computational means of enhancing their research and the resulting narratives, such as geographic information systems (GIS) and social network analysis (SNA).

Modeling in the natural sciences, cooperation studies, and history

Scientists develop mathematical models for many different reasons, which span both the theoretical and the practical. Within the more theoretical realm, they create models to represent and help explain a physical principle or process, to test and revise theory, and for hypothesis testing. Perhaps the most famous mathematical model in the world is Albert Einsteinôs $e = mc^2$, which serves to represent and predict the relationship between energy, mass, and the speed of light. When most natural scientists think of models, however, they typically do not think of a single equation but envision an array of mathematical equations that represent a number of physically-based principles or processes, often coded into a computer program, which are collectively used to simulate the system of interest.

Models typically are not used for purely theoretical purposes; they are used to perform practical tasks. Challenging situations sometimes present themselves: the system is not well understood and/or the data are ambiguous, contradictory, and otherwise so messy that interpretation becomes difficult. Here, models may still provide useful insights into underlying complex system dynamics and even provide reasonable prediction capability. Models can help satisfy a number of important objectives, including: (1) Providing a theoretical framework for a system of interest. (2) As a means for testing and assessing a hypothesis. (3) By highlighting what is not yet known, stimulating the formulation of the most important remaining research questions to organize the search for further evidence. (4) Giving insights into system behavior. (5) Offering useful predictions of future behavior or outcomes.

Models are used in many different disciplines in the natural sciences, engineering, and business. They have been developed for a vast array of applications, ranging from the familiar to

the esoteric. Familiar applications include stock market price movement, weather prediction, and traffic flow analysis. Work on flight patterns of birds, water movement through soil, and snow avalanche analysis offer more esoteric examples. In these practical applications, new models or theories are rarely developed. More often, researchers employ existing models whose physical and mathematical foundations are considered appropriate for the system and prediction problem of interest. Professional software programs that implement industry standard models in a user-friendly environment are widespread.

However, mathematical models are rarely used in other fields. These fields are largely within the so-called ñsoftò disciplines of the humanities, which include history and literature. There are notable exceptions, particularly with the social sciences, in which human interactions and decision-making are sometimes modeled with mathematical techniques such as game theory and agent-based modeling (for examples from cooperation studies, Castelfranchi and Falcone 2010; Gintis 2009; Han 2013; Sigmund 2010).

Within the historical sciences, models have generally only begun to be used within the last twenty years, and generally, they exhibit limited scopes and applications.[2] Historians often lack the data for both developing and testing models, which constitutes one of the obstacles for using traditional mathematical modeling approaches. In addition, historical actions, events, and processes are not subject to explicit representation by physical laws but rather involve human behavior, which is inherently difficult to model because the observation and reporting of such behavior are always imprecise. Although there are mathematical models that have been developed for this purpose, including game theory and agent-based modeling, they are not subject to the same rigor and foundational basis as models of more physically based problems.

A further complication is the nature of historical data, which are often fragmented, incomplete, vague, and inconsistent. These information challenges make it even more difficult to attempt to model historical systems that are already hard to conceptualize and represent with a mathematical formulation. Often, solid data do not exist, and the modeler is forced to assess subjectively the factors that she or he believes describe the phenomenon correctly. Interestingly, this subjectivity applies even to the so-called ñhardò sciences. As pointed out by historian of science Naomi Oreskes and her collaborators in their classic 1994 paper, ñFundamentally, the reason for modeling is a lack of full access, either in time or space, to the phenomenon of interestò (644). If the model accurately predicts the outcomes under a range of conditions and/or time periods, including outcomes in the past that are subsequently discovered by historians, this prediction provides some degree of confidence that the theory and the corresponding model offer a useful representation of the phenomenon, which can be used to challenge existing ideas.

This article introduces a simple fuzzy rule-based model for a real-world social cooperation case. The model is highly simplistic and is not grounded in any formal theory of cooperation. Martin Nowak (2006) places the existing forms for research on cooperation into the

[2] Modeling in the historical sciences includes Fogel and Engerman 1995 (1974); Greif 2006; Korotayev et al. 2015; Preiser-Kapeller and Daim, eds. 2015; Rosser 2000; and Weidlich 2000.

categories of *kin selection*, *direct reciprocity* (perhaps similar to the way that historians approach patron-client relationships), *indirect reciprocity* (which may be important for modeling human cooperation because *reputation* is an important factor), *network reciprocity* (where populations are not well-mixed and one identifies network clusters), and *group selection/multilevel selection* (where a group of cooperators is more successful than a group of defectors from cooperative relationships). These categories are themselves subjective, and the network matrices require assumed values. The models created by researchers working within the first three of Nowakôs ñrulesò will be difficult for historians to apply because they assume the existence of ñwell-mixedò populations, which historians may not find in their sources. Using a focus on cooperation, we wish to apprise readers of the possibilities for applying fuzzy rule-based models to historical problems using a fuzzy logic linguistic approach.

Description of the social cooperation dilemma

In a cooperative situation, an individual behaves in a way that is costly to the individual but provides a ñpublic goodò (net gain) that benefits other individuals. Cooperation poses a dilemma because, in a cooperative community, the individual who does not cooperate or whose contribution is lower than that of others will often derive a higher benefit, as for that individual, the value of his or her cost of cooperation will be lower. If the ñcheaterò, ñfree-riderò, or ñdefectorò always wins, why should an individual cooperate? This dilemma should eventually undermine cooperation among humans and lead to the destruction of ñpublic goodsò, such as fisheries, grasslands, clean air and water, and organized workersô economic and political influence (the classic theoretical work is Hardin 1968; see [2009 Nobel laureate] Ostrom 1990). Yet, despite potential harm and destabilization caused by those who do not fulfill their obligations, cooperation is ubiquitous in human communities and throughout nature (as pointed out by Kropotkin 1902). How is this ubiquity possible? We started thinking about modeling historic cases of cooperation using fuzzy logic or fuzzy set theory as part of our research project for the European Science Foundation program ñThe Evolution of Cooperation and Tradingò (TECT), which owed its existence heavily to evolutionary biologists and economists, whose classical theories focus on *competition* (Owens 2011, 2012; for overviews of cooperation research, see [TECT leader] Noë 2010; Nowak and Highfield 2011).

Cooperation exists not only among humans but between human beings and other species (and between nonhuman animal and plant species: Bronstein 2009; Bshary and Bronstein 2004). Of these, perhaps the shepherd and the livestock dog provide the best known example, where a guardian dog cooperatively works with the shepherd to protect vulnerable livestock from predators (Range and Virányi 2015).

We study a complicated social cooperative network in Twofold Bay, southeastern Australia, over a century until 1930. Surviving sources document that ñpodsò of killer whales or orcas (*Orcinus orca*) worked cooperatively with human bay whalers. Most people in Twofold Bay live in the small town of Eden, whose name was entirely too enticing for the creators of the

PBS television documentary ñKillers in Edenò (Toft 2004). The cooperation is extensively documented in Thawa Aboriginal oral histories, European oral accounts, including filmed interviews with elderly witnesses for the PBS documentary, diaries, official reports, newspaper articles, photographs, and reportedly an old movie that disappeared. However, the whalers are long dead, and there is no systematic collection of records from which we can draw. For modeling, we also have important studies of orca behavior in different regions of the world done over the past 40 years and significant research on Australiaôs Aboriginal peoples (Clode 2011). In the absence of hard data for the Twofold Bay case, simple fuzzy heuristic rules can be used to construct a basic model that simulates this complicated social network behavior.

In the second half of the period of shore-based, bay whaling in Twofold Bay, several of the orca pods reportedly grew close to one group of whalers. The most dramatic and perhaps sensational accounts purport that as the relationship developed, the ñkillersò (as called by their human collaborators) would alert the whalers when baleen whales (Mysticeti) were passing on their migration route between Antarctica and warmer eastern Australian waters. Orcas swam to the coast of the bay to signal the whalers by flopping themselves on the waterôs surface to make noise. The orcas, sometimes with as many as three pods cooperating with each other, used their numerical advantage to target and attack a much larger baleen whale, driving it from deep water into the shallower bay where it could not dive out of reach and, if necessary, blocking its escape. By attacking and exhausting the animal in often far-flung battles around the bay, the killer whales rendered the baleen whales far less dangerous to the human whalers in their small, open boats.

When a large baleen whale was successfully killed by this cooperative network of humans and orcas, the human whalers, apparently on the advice of the Aboriginal whalers among them, rewarded the killer whales with the first access to the dead baleen. An Aboriginal story attributes the beginning of human-orca cooperation to an elderly man who, prior to European arrival, got the help of the killer whales to drive other species to shore. During the later cooperation with bay whalers, the orcas would pull the baleen carcass into the depth of the bay, where they would gorge themselves on the body parts they preferred, primarily the tongue and lips. When the killer whales finished consuming their share, the carcass would float to the surface within a couple of days, with a marker of the successful human whalers attached, and the humans then towed the carcass to the tryworks where it would be processed to obtain its oil, bone, and meat. By the time when photographs were taken, these humans used almost exclusively the tryworks at the mouth of the Kiah River, across the bay from the municipality of Eden. The Europeans called the simple rule for the division of the spoils, the ñLaw of the Tongue.ò

Ultimately, outside human agents destroyed this *trust* between the killers and the human whalers. A wandering European murdered a beached orca, who was an important part of one of the remaining pods, in sight of the other killer whales and many Aboriginals. The Aboriginals apparently believed that as totem kin, the souls of some of their ancestors had passed to orcas newly arrived in the bay, and they often named these new killer whales for recently deceased

Aboriginal whalers. Sometime after the murder, all of the Aboriginals struck their long-time ñcampò and left the Twofold Bay area, depriving the Europeans of an important part of their knowledgeable whaling workforce, which was especially trusted by the orcas. New arrivals to the whaling business used loud, gunpowder harpoons, which likely disrupted the sound communications of the killer whales, and these newcomers sometimes violated the ñLaw of the Tongue.ò Few orcas returned to the bay following the murder and the departure of the Aboriginals, and although a level of cooperation remained for about fifteen years with these few, until the death of the last of them in 1930, the social cooperative network disappeared and was never revived. Whaling in Twofold Bay had died, and in about twenty years, large, ocean-going whaling ships, which had already begun working in the baleen migration route by the early twentieth century, would slaughter the baleen species and reduce the populations of many of them to levels that threatened them with extinction.[3]

The fuzzy rule-based modeling problem

Our goal is to develop a fuzzy rule-based model that predicts the likelihood of the success of the social network in killing a whale. For this historic case, ñpredictionò means that the model will simulate the outcome of a whale hunt for each event in our narrative. For this purpose we employ *fuzzy set theory*.

Now a well-established mathematical technique, fuzzy set theory (or *fuzzy logic*) arose from the observation by Lotfi A. Zadeh (Computer Science Division, University of California-Berkeley) that the real world is often *vague*, with imprecise boundaries or classifications between entities (Zadeh 1965, 1973, 1976, 1997; on fuzzy set theory and vagueness, see Novák 2006a, 2006b, 2012 [especially p. 42], Novák and DvoŚák, forthcoming). Fuzzy logic makes use of *linguistic variables* whose values are words, phrases, and sentences in natural language. Fuzzy rule-based modeling requires of a set of rules, which consist of linguistic ñif-thenò premises that attempt to capture the fundamental system variables and conditions that govern the behavior of interest. The ñifò represents the inputs or predictor variables that collectively estimate or predict the ñthenò or output variable(s) that represent the system behavior or output of interest. By substituting fuzzy numbers that represent the variables within the ñif-thenò premises, the linguistic rules are transformed into a fuzzy mathematical rule structure that captures the inherent vagueness of the system (Zadeh 1972, 1975). For a given system condition or event, the set of input values are mathematically processed through the fuzzy premises from which the applicable fuzzy rules for the given event are identified. The identified rules are then combined

[3] We pieced together this account from Bosa 2009, 2010, 2012; Clode 2011; Crew 2014; Eden Killer Whale Museum 2013; Jones forthcoming; McKee, n.d.; Mead 1992; NSW National Parks and Wildlife Service 2014; Pearson 1985; Russell 2012; Staniforth 2008; and Toft 2004. We benefited from the discussions of Matthews 1904 and Wesson 2000 by Clode 2011.

proportionately to the degree to which *domain experts* (historians, in our specific case) think they apply, and they are then ñdefuzzifiedò to obtain a single crisp answer or output value.

As a fuzzy set is a set of objects without clear boundaries, fuzzy sets are characterized by having varying *degrees of membership* to a particular class. The transition from membership to non-membership is gradual, a continuum of values. For example, cooperation between orcas and human whalers may be divided into different classes, which could include low (or weak) cooperation, medium (or moderate) cooperation, and high (or strong) cooperation, where ñmediumò denotes a degree of cooperation that in general exceeds low cooperation but is less than high cooperation. In some instances, there may be overlap between classes. If these ñclassesò corresponded to the fields of a spreadsheet or other table, we might still feel that some of our observations should be represented in more than one field, but in creating a simple model, we would likely wish to limit the number of classes. Fuzzy membership functions can represent qualitative (e.g., cooperation) or quantitative variables (e.g., temperature).

Mathematically, a fuzzy number ascribes vagueness to a variable through its membership function. Membership to a class ranges from 0 to 1, with 0 representing no membership and 1 complete membership. The values in-between show degrees of partial membership to the class. An example would be classifying *moderate* temperature (a vague concept) expressed as a *triangular fuzzy number* (TFN). A TFN represents an imprecise value or set of connected possible values, which constitutes a range around the central value (the high point of the triangle) and minimum and maximum values of degree of membership on either side (which form angles with the ñXò axis). In our example, we could assign values ranging from 45 degrees to 65 degrees Fahrenheit. For this class, a membership of 0 (i.e., no membership) occurs at both of these two extreme points; complete membership occurs at 55 degrees; and varying degrees of membership to the moderate temperature class occurring between this midpoint and the extreme points. Fuzzy numbers can assume different geometric shapes, but triangular numbers are the most common in this sort of work.

The model, like any model, can be made far more complicated through the addition of more variables, classes, and rules, and/or by combining the model with other modeling techniques, such as ñagent-basedò modeling or evolutionary set theory (Smithson 2005; Tarnita et al. 2009; Zadeh 1994a, 1996). The relationship between fuzzy set theory, as a technique to deal with vagueness, and probability theory, as a technique to deal with uncertainty when the statistical universe is known, remains controversial (Novák and DvoŠák, forthcoming; Verkuilen 2005: 468-70; Zadeh 1994b, 1995, 2004, and 2014). It should be mentioned, however, that greater model complication does not necessarily yield a more accurate or complete representation of something observed in the real world. In fact, if a model attained completeness, it would no longer be a model. All models are simplifications, often gross simplifications. The world is complicated; our models do not have to be complicated because simple models have their uses. We can use them to note difficulties with other approximations of the situation of interest, and models enable us to spot understudied aspects of a system or where additional empirical data are needed. Beyond their heuristic function, models cannot be ñprovedò in the

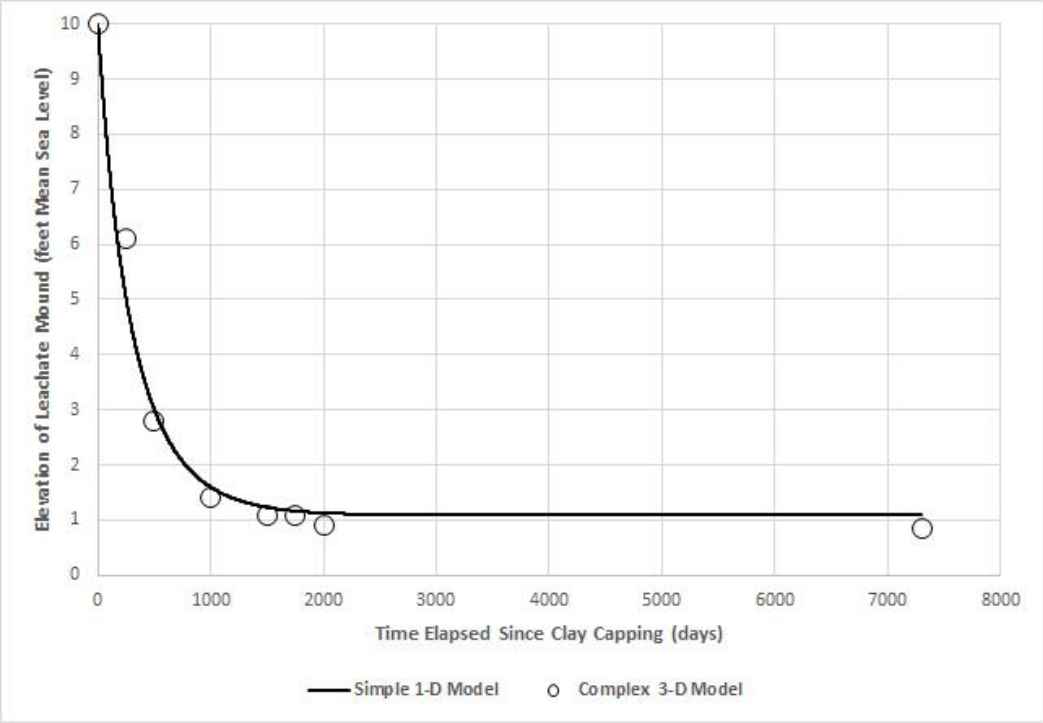
usual sense of that term (Oreskes et al. 1994). Furthermore, we are often forced to make simplifying assumptions that we suspect diverge from anything in the real world in order to engage in observation or to make measurements. Historians make the same sorts of assumption in creating the approximations and abstractions they present in their articles, books, and maps. Our knowledge is always incomplete. Yet, even simple models often provide us with some insights into the complex system, and sometimes with a simulation/prediction capability. If nothing else, they force us to formulate a formal structure of the system, which we can assess, revise, or reject in favor of some improved formulation.

To offer a classic example, the climate change modelers in the early 1970s started with extremely simple models, and they knew these models were laughably simplistic from what they needed to become: for example, using highly simplistic one-dimensional models to simulate heating and cooling of our three-dimensional planet. Interestingly, even as recently as the 1970s in field as rigorously quantitative as climate science, Stephen Schneider, universally revered as a major contributor to climate modeling, observed that ñcomputer models back then were regarded with suspicion by traditional empirical scientistsò (Schneider 2009: 37). Still, these early models provided scientists with useful and non-obvious insights and reasonably accurate correspondence to the data they had, which allowed them systematically to build more complicated models that incorporated more features that more closely approximated the real-world processes. These simple models, which Schneider described as ñback of the envelopeô models because they may be only one equation long, which could be scribbled on the back of an envelopeò (36), finally evolved into the complicated three-dimensional global circulation models used today. These more complicated models have not changed the basic lesson, gained from their simpler predecessors, about the role of human behavior in stimulating climate change (AIP 2015a, 2015b; Manabe and Wetherald 1975; Weart 2003-2015, 2008).

Even very simple models can sometimes achieve the equivalent predictive accuracy as a far more complex model. For example, as depicted by Figure 1, Coppola et al. (2002) developed a relatively simple 1-D water balance model that achieved almost the same accuracy as an advanced 3-D numerical model developed by another group for predicting declining leachate (i.e., water) levels at a large municipal landfill. The more advanced model utilized tens-of-thousands of equations representing multiple geologic layers and varying boundary and initial conditions (e.g., surface water levels of nearby water bodies). In contrast, the far simpler 1-D model consisted of one equation and in effect, averaged the various geologic layers and conditions into single representative values. By extension, while the simple 1-D model required minimal data inputs and was executed in a simple Excel spreadsheet, the complex 3-D model required many tens-of-thousands of data inputs and was solved using advanced numerical algorithms. However, due to the fact that the far simple 1-D model was able accurately to represent and simulate the predominant physical processes, it was able to reproduce almost exactly the predictions provided by the advanced numerical model over a 20-year forecast period. It should also be mentioned that the same simple, 1-D model was able to reproduce

accurately actual measured leachate levels at a municipal landfill over a 40-year period during both its growth and dissipation.

Figure 1. A Comparison between Predictions Generated by a Simple 1-D Spreadsheet Model and a Complex 3-D Numerical Model



We provide this physical world example to emphasize to historians that model simplicity does not necessarily imply a lack of rigor, sophistication, or even predictive capability (i.e., accuracy). Unfortunately, for some mathematical modelers, there is a pervasive, stubborn deference to advanced models over simpler ones, as if increasing model complication is a necessary condition for establishing its validity, improving prediction capability, and providing useful insights.

This complaint is not to imply that ignoring fundamental processes for the purpose of constructing a simpler model is preferable. We recognize that more complicated models also rely on simplifying assumptions, which often fall short of representing the real-world system, and by extension, often require significantly more data inputs that may be highly uncertain. As a matter of practical consideration, even when a more complex model is needed, it is often of great benefit to begin with a simpler model to test basic assumptions and data, and from this test, more advanced models can follow, which is how climate models evolved.

One of the inherent advantages of fuzzy rule-based modeling is that models can range from the very simple, consisting of few premises and rules, to the highly complicated, consisting

of many premises and rules. This article does not present the methodology of fuzzy rule-based modeling in detail, but rather, the general concepts and approach.[4]

Modeling approach

For this modeling exercise, we developed a simple fuzzy ruled-based model, which simulates the cooperation between the orcas and the human whalers and provides their likelihood for success for a particular hunt given the degrees of cooperation and opportunity. For this model, we used two predictor variables: cooperation and opportunity. Cooperation entailed trust between the orcas and human whalers. It might also imply the presence of particular leaders for each group. For example, ñOld Tomò was a large, venerable, and successful killer whale, who appeared to play a leadership role even though orca pods are usually led by females. Tom developed a strong relationship with George Davidson, the last of the Twofold Bay whalers. With Tomôs death in 1930, all orca-human cooperation ended. The category of cooperation could also include the number of killer whales, their fitness level for predatory success (e.g., adults vs. juveniles), and the levels of trust between the Aboriginal and European whalers.

A number of variables that pertain to the prey, the hunters (their knowledge and skill), technology, and natural conditions define opportunity. This prediction variable could include proximity of the baleen whales to the hunting area, the learning curve of the baleens about how to resist and save themselves, the number of prey, the presence of baleen calves, the number of orcas and human whalers, their fitness level for predatory success, and weather conditions (hunting took place in the winter from June to October). Rather than lump these factors into ñglobalò variables of cooperation and opportunity, these additional variables could be individually represented and used to construct a more complicated model, but that would require more data and/or domain expertise.

One can see overlap between these two variables: for example, predatory fitness could be used to characterize cooperation (i.e., how effectively the human whalers and orcas work together) which in turn promotes opportunity. In addition, the ñupdatingò of this opportunity, in the sense used in Artificial Intelligence (AI) agent-based modeling, encourages the participants to cooperate to achieve a successful hunt. This overlapping provides another level of fuzziness as we define the ñenvironmentò in which all of the actors were involved and we sort out the essential characteristics of the story we might wish to tell. Here, we wish to offer historians a framework for a discussion of a valuable case study in cooperation, from which they can learn about fuzzy rule-based modeling.

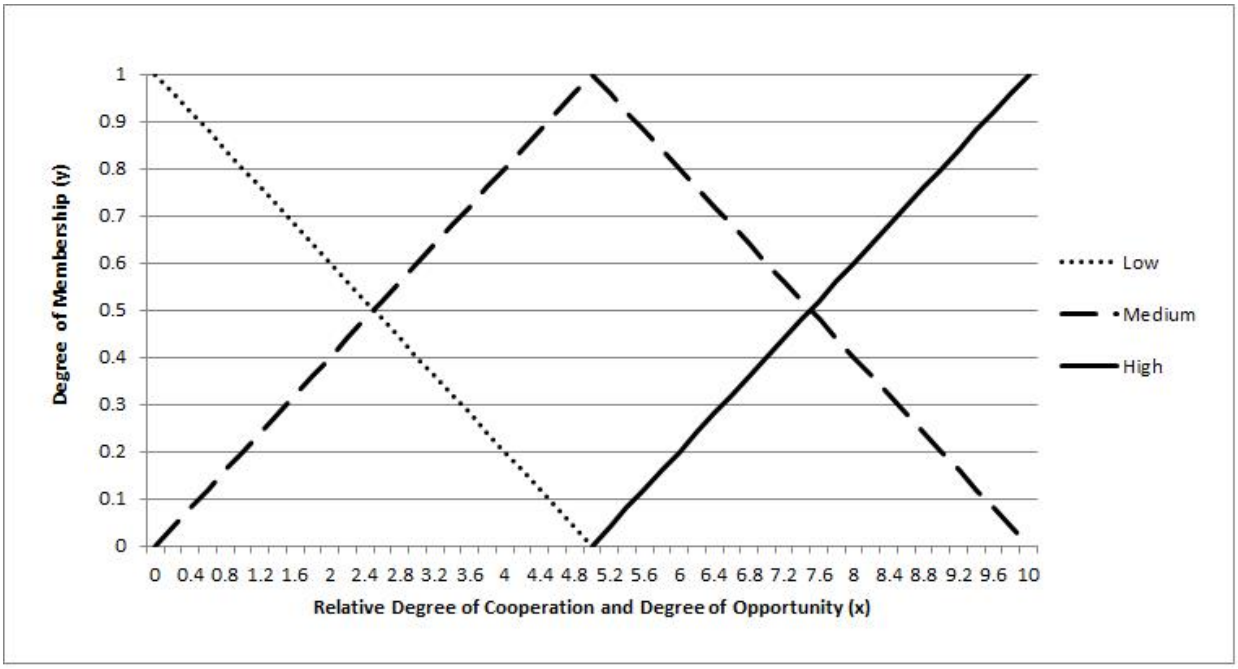
[4] A far more detailed treatment with specific examples may be found in Coppola et al. 2002. For a free, online tutorial, see Kaehler 1998. For a defense of the application of fuzzy set theory in the social sciences, see Smithson and Verkuilen 2006.

Variables, classes, and rules defined

As presented above, two input variables, *cooperation* and *opportunity*, are used as predictors to estimate the likelihood that the social network of orcas and human whalers will successfully kill a baleen whale. Both variables were classified into three classes: low, medium, and high. These classes were arranged along an arbitrary, relative scale ranging from 0 to 10, with 0 denoting the minimum possible value and 10 the maximum possible value for each variable. In this example, we assigned both cooperation and opportunity the same fuzzy numbers for each of the three membership classes. In practice, these two variables do not have to share the same classes and/or fuzzy numbers. Obviously for variables that represent different characteristics or physical measures (e.g., temperature versus precipitation), the classes and fuzzy numbers will often be different, even within a single fuzzy-rule set. The number of classes and fuzzy numbers can be determined using the best judgment of domain experts and/or formal mathematical algorithms. The presentation in articles, books, or online of these classes and the fuzzy numbers developed in the modeling process will permit productive discussions among the various domain experts, which will assist in refining all aspects of the resulting model. We feel that improving the quality of academic debates will be an outcome of the use of fuzzy rule-based modeling, which will enhance the quality of History and related disciplines involved.

Figure 2 shows the fuzzy numbers for these two input variables and three membership classes. Note that the Y axis denotes the degree of membership that a particular class has with respect to its relative degree of cooperation and opportunity. As can be seen, the medium class overlaps with both the low and high classes. It is within these zones of the graph that the degree of membership is truly ñfuzzyò.

Figure 2. Cooperation and Opportunity Membership Functions for the Three Classes



Each combination of the different variable classes produces a different rule, for a total of nine possible *linguistic rules* (three different classes for each of the two different input variables).

Table 1 summarizes the different possible fuzzy classes for each variable and corresponding linguistic rule outcome.

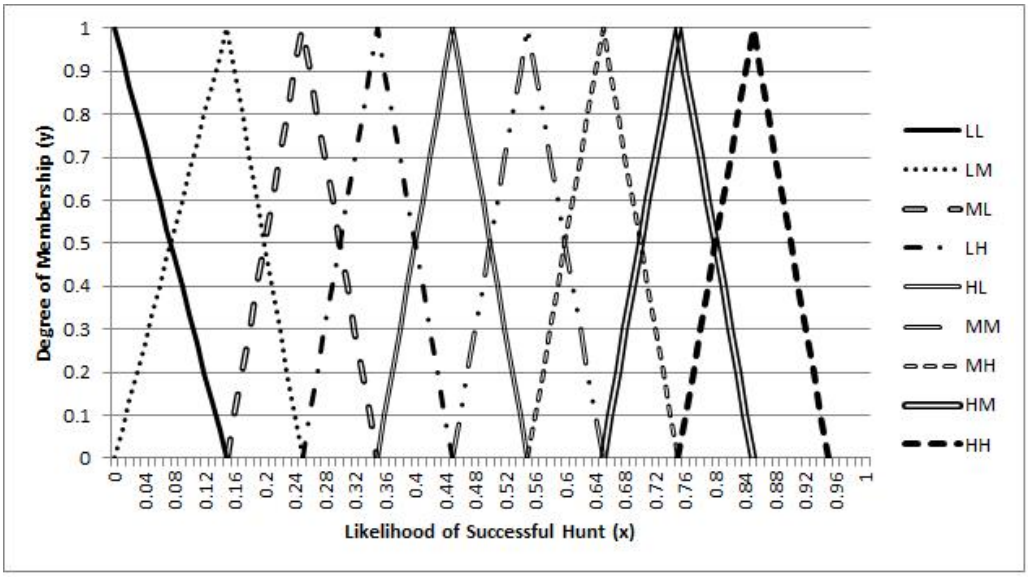
Table 1. Set of Linguistic Rules for the Two Variables and Three Classes

Cooperation	**Opportunity**	**Fuzzy Rule Outcome**
Low	Low	LL
Low	Medium	LM
Medium	Low	ML
Low	High	LH
High	Low	HL
Medium	Medium	MM
Medium	High	MH
High	Medium	HM
High	High	HH

For example, the first rule states that ñIf cooperation is low and opportunity is low, then the likelihood for hunting success is some fuzzy number LL.ò This fuzzy output number representing the possibility of hunting success must similarly be developed based upon the available information, or in this case, domain-expert judgment. In other cases, when sufficient data exist, learning algorithms can be used to derive the fuzzy rules. Coppola et al. (2002) present a more detailed description of a formal fuzzy learning algorithm and how it is applied to real-world data. In this case, because of the lack of sufficient information, domain-expert judgment was used to formulate these rules.

Figure 3 shows the actual fuzzy likelihood numbers that correspond to each of the nine fuzzy rules.

Figure 3. Likelihood Membership Functions for the Nine-Rule Set



The y axis denotes the degree of membership, and the x axis denotes the likelihood of hunting success. As Figure 2 shows, the fuzzy rule that has both low cooperation and low opportunity provides a declining membership function, with likelihood of success ranging from 0 to less than 0.15 (reaches a 0 membership at likelihood of success of just 0.15). All other fuzzy-rule outputs are triangular fuzzy numbers (TFN). For example, at the other extreme, the fuzzy rule that has both high cooperation and high opportunity has a likelihood range of success between 0.75 and 0.95, with an average value of 0.85.

As the fuzzy response numbers show, unlike the inputs, the rules are not symmetrical in nature. In this case, we gave slightly more weight to cooperation than opportunity in terms of the possibility for hunting success. Stated another way, we assumed *a priori* that the likelihood of a successful hunt is more enhanced by cooperation between humans and orcas than by opportunity because the available information suggests to us that this relationship continued to exist until outside entities altered key variables.

We assumed this enhancement without sufficient supporting empirical evidence. This conclusion may or may not approximate the actual situation, or there may have been circumstances where opportunity rather than cooperation was more important for increasing or decreasing the likelihood of hunting success. For example, factory ships operating around 1900 in the baleen migration routes off the coast may have so altered the opportunity side that

cooperation dissolved and the success of the bay whalers declined to the point where this activity could not be sustained. There may even be a threshold where the relative importance of the two input variables reverses. That is, below a certain level of cooperation, it may be better to have more opportunity, but above a certain level of opportunity, more cooperation may become more important. Without more complete information or a deep theoretical foundation or understanding, the modeler, as is common in historical studies, is forced to make assumptions that may or may not be well-founded. The advantage of modeling is that one can always try different alternative, competing models and reject, accept, or modify as the results dictate or as more data and understanding emerge. *Because the vagueness of the problem can be captured within a formal modeling structure for testing and assessment, fuzzy rule-based modeling offers historians and other cooperation researchers a significant advantage over other forms of inference.*

Fuzzy output generation

Using a method described further below, Table 2 lists the outcome for different events, where different numbers pertaining to cooperation and opportunity were selected. From this process, the relevant fuzzy rules were identified, combined, and then ñdefuzzifiedò to yield a single ñcrispò value of likely success.

Table 2. Sample Outcomes Using Fuzzy Rules

Cooperation Input Value	Opportunity Input Value	Crisp Success Likelihood
0	0	0.075
0	1	0.09
1	0	0.11
2	2	0.2
3	2	0.26
2	5	0.3
3	4	0.34
5	2	0.37
2	7	0.37
3	6	0.41
7	2	0.45
6	3	0.47
2	10	0.47
5	5	0.55
10	2	0.57
5	7	0.59
7	5	0.63
7	8	0.69
8	8	0.73
8	9	0.75
9	8	0.77
10	10	0.85

As the table shows, even with a 0 assigned input value to both cooperation and opportunity, the computed likelihood of success for the defined fuzzy rule model is 0.075. Actually, this outcome is not credible when weighed against what we do know about shore-based, bay whaling, in Twofold Bay and other places where it was practiced. However, it reflects that the inputs are ñfuzzyò or imprecise, and hence, there may always be some possibility of success, even under the most challenging circumstances. Alternatively, this rule outcome could be modified to yield a 0 outcome value. At the other extreme, because the existence of both perfect (i.e., 10) cooperation and opportunity does not guarantee success, its likelihood must be less than 1.0 (in this case, 0.85).

In order to illustrate how one performs rule combination and defuzzification, we provide two examples. As presented in Table 3, for the first example computation, cooperation and opportunity values of 6 and 3 were assumed, respectively. We then computed the corresponding membership function values for both of the two paired variables for each rule. The product of these two membership function values is the *degree of fulfillment* (DOF), which is a measure how strongly a particular rule applies to the particular event. We deemed only those rules with a minimum assigned DOF threshold of 0.05 to have sufficiently high membership to the rule. For this DOF threshold, four of the nine possible rules apply for this particular case: Medium Cooperation and Low Opportunity (ML); High Cooperation and Low Opportunity (HL); Medium Cooperation and Medium Opportunity (MM); and High Cooperation and Medium Opportunity (HM).

Tables 3 and 4 present the applicable rules for their cases, the corresponding degree of membership to the particular class for each fuzzy variable (e.g. 0.8 degree of membership of cooperation for medium membership class in the rule ML), and the computed DOF as the product of the two degrees of membership. M(P) denotes the mean success prospect value for each rule (e.g., medium cooperation and low opportunity has a mean success possibility of 0.25). The product of each DOF and M(P) is computed for each applicable rule. We performed defuzzification by combining the M(P) values for the four rules proportional to their DOF, yielding a single, *crisp*, success likelihood value of 0.47 for the first case. In effect, defuzzification computes the weighted average of the mean output values for all relevant fuzzy rules, proportional to their DOF.

Table 3. Rule Identification, Combination, and Defuzzification for Cooperation = 6 and Opportunity = 3 for a Minimum Degree of Fulfillment (DOF) of 0.05

Applicable Fuzzy Rule	Cooperation Degree of Membership	Opportunity Degree of Membership	Degree of Fulfillment (DOF)	Mean Prospect M(P)	DOF x M(P)
ML	0.8	0.4	0.32	0.25	0.08
HL	0.2	0.4	0.08	0.45	0.036
MM	0.8	0.6	0.48	0.55	0.264
HM	0.2	0.6	0.12	0.75	0.09
Defuzzified Likelihood					0.47

For the second case, we reverse the values assigned to cooperation and opportunity, with cooperation assigned a value of 3 and opportunity a value of 6.

Table 4. Rule Identification, Combination, and Defuzzification for Cooperation = 3 and Opportunity = 6 for a Minimum Degree of Fulfillment (DOF) of 0.05

Applicable Fuzzy Rule	Cooperation Degree of Membership	Opportunity Degree of Membership	Degree of Fulfillment (DOF)	Mean Prospect M(P)	DOF x M(P)
LM	0.4	0.8	0.32	0.1333	0.042656
MM	0.6	0.8	0.48	0.55	0.264
LH	0.4	0.2	0.08	0.35	0.028
MH	0.6	0.2	0.12	0.65	0.078
Defuzzified Likelihood			1		0.41

From these two case examples, we illustrate the non-symmetrical values of cooperation and opportunity, with the importance of cooperation between human whalers and orcas emerging as more important than opportunity in achieving hunting success for this particular fuzzy rule-based model. That is, reversing the value of the original input values of 6 and 3 assigned to

cooperation and opportunity, respectively, reduces the probability of hunting success from 0.46 to 0.41.

As shown here, in many cases, four fuzzy rules apply to a particular event or set of input values. In rare cases, just one fuzzy rule would apply. For example, if we assigned both cooperation and opportunity a value of 5, then the only fuzzy rule would be medium cooperation and medium opportunity. Note that non-whole numbers could be used as inputs. For example, a domain-expert may wish to estimate cooperation as 5.2 and opportunity as 4.7, and the same methodology would apply. In the examples we offer, we use whole numbers to illustrate the method, with the recognition that for such a complicated and vague problem, a domain-expert may not find it possible to distinguish between 5 and 5.2. Again, a more detailed description of this method may be found in Coppola et al. (2002).

Conclusions

In this article, we present a simple, fuzzy rule-based model we created of a cooperative social network between shore-based human whalers and killer whales (*orcas*) to hunt large baleen whales. Using cooperation and opportunity, we developed fuzzy rules to estimate the prospect of hunting success based upon cooperation and opportunity. This example demonstrates how the rule outcomes can be derived without explicit quantitative data by using domain-expert judgment. In the presence of a database of numbers (for example, quantifying the whalers, the orcas, and their catch by month or year), we would derive the rule outcomes by using a learning algorithm (reasoning mechanism).

We employ the strengths of fuzzy logic because with this well-established mathematical technique, data can be imprecise or vague and we can use variables that are subjective in nature (for example, *cooperation*). The rules are transparent and are thus readily understood and discussed among domain experts and with prior success, some of the same rules may be used in many other areas of engineering and science. Rules can establish boundaries for the expected range of system behavior under different conditions by defining unlikely situations (for example, ñIf drought is severe, then corn prices will be lowò). The modeling process remains open-ended so that in the future, we can include other models and/or variables/classes, and we can perform agent-based modeling to simulate the behavior of the agents over time (as in Han 2013), thereby capturing the historical dynamics of the narrative within the structure defined by the original simple model.

This example also illustrates how lumping several variables into one (for example, weather and number of whales into opportunity), while providing the advantage of simplicity, has limitations. However, for an initial problem formulation, this example is useful. It also illustrates and demonstrates the possibilities for modeling historical systems that are vague and for which the surviving information is imprecise. And for the purpose of illustration and demonstrating the possibilities for modeling historical systems that are vague and for which the surviving information is imprecise, this example serves its purpose. If one could construct a time machine and go back to Twofold Bay in the nineteenth century or the early part of last century

and interview the whalers, or better yet, accompany them on their hunts of baleens over a number of years and accurately record parameters of the hunt (for example, location of the baleens, number and type of baleens, the direction of baleen movement in the bay, weather and ocean conditions, the number and condition of the orcas, all stored in a database), a more rigorously defined fuzzy rule-based model could be developed. However, much of our information would still be vague because it would come from observations underpinned by informed judgment and convenience to make observations possible. For example, Clode (2011) sheds more light on the Twofold Bay cooperation through a comparative study of orca pods in different regions of the world, which could be used to enhance the simple model proposed here or to create one or more competing models. These other orca pods have identified different prey, and they have often drastically different hunting techniques from those observed in Twofold Bay. If their numbers are not decimated by human predation or environmental disaster, all whale species in our model are intelligent, social animals who learn, alter their behavior, and pass essential information to subsequent generations (Jones forthcoming). Without the ability to use a shallow bay in cooperation with shore-based human whalers, and in the presence of the devastation of baleen numbers that followed the increased use of factory ships, did orcas along the coast of southeastern Australia continue to hunt baleens? The orca pods in that zone may have had to identify other prey or alter their hunting techniques, and orcas are clearly able to do both. Moreover, our knowledge about a number of variables would remain incomplete, ambiguous, and contradictory: for example, the *trust* between the orcas and human whalers, which appears to have been a fundamental aspect of the narrative at the beginning of the twentieth century.

As with many if not most historical problems, we are left with anecdotal and often conflicting accounts and sparse data, and from these fragments of information, we constructed a simple fuzzy rule-based model. Our attempt to represent a complicated social network with a simple rule structure falls far short of plausibility. At the same time, our initial efforts, however modest, compel the historian/modeler to formulate a set of linguistic rules that quantify often highly vague variables and conditions, qualitative and/or quantitative in nature, in an attempt to represent and simulate a complicated system of interest. As we learn from our initial models and accumulate more data, information, and understanding, we can formulate and test new models against the surviving record, allowing us to consider alternative hypotheses and to see more clearly what additional information we need to acquire (for example, about the social and cultural environments of the Aboriginal groups of southeastern Australia) in an attempt to explain better this fascinating cooperation between orcas and human whalers for at least a century to hunt successfully large baleen whales.

As stated at the outset, the historical model application presented here is more pedagogical in nature. We present the methodology for constructing a simple fuzzy model for a vague but complicated social cooperative network, along with example model-simulation results. In terms of the usefulness of the actual outcomes generated by this extremely simple model to the real-world cooperative relationship between the killers and the whalers, it is impossible to

assess the results for this particular prediction problem until experts on shore-based whaling in Twofold Bay publish additional information. For example, is cooperation indeed more important than opportunity? Does the relative importance of these two variables change under certain conditions? Does hunting success promote further cooperation among the two species? Even with further research, it may be impossible to assess accurately the model results with existing data, or at least data that have already been systematically chronicled and organized. Still, the fuzzy model presented here provides a formal framework, which can easily be modified to represent more accurately the available information.

Using available data, which may be largely anecdotal in nature, the historian can tabulate failures and successes for distinct hunting events based upon these two variables. As stated earlier, each variable could entail a sub-set of variables; for example, opportunity could include the number of baleen whales and ages (e.g., calves), the weather conditions, proximity to the coast, with the outcome of the hunt noted for each event. This type of systematic data compilation in combination with the preliminary fuzzy rule-based models can ultimately be used not only to extract more information but to further refine the model, create new models with more variables and rules, and test their predictive accuracy against the known historic events. In effect, there is a feedback process, where the model helps to drive the data assimilation and analysis process, which in turn is used iteratively to refine and test the model(s). The results of this process will then enhance the clarity of the historical narrative we need in order to understand adequately the dynamic system of which the orca-human cooperation was a part.

To use the previous example of climate modeling, early simple models were responsible for identifying ñcloud feedbackò as an important factor that affects warming and cooling of the earth. This insight drove future modeling and data collection efforts, which in turn lead to further improvements in both. Using this particular scientific breakthrough as a simile, most modeling problems, whether in the physical sciences, and perhaps particularly in the discipline of History, are impaired by imperfect information and our flawed understanding of processes that are often so complicated that no model will ever accurately capture the underlying dynamics, which is why narratives are so necessary (Allen and Hoekstra 2015: chap. 8). Many models will likely fail to achieve any semblance of acceptable prediction accuracy. Still, however seemingly simple, models often serve as the first step to forming a theoretical or heuristic framework for analysis, which can be used to refine and improve our understanding through additional data collection, model development, and testing.

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